

BISOCA, Romania

WMO: 152850

Lat: 45.5489N

Lon: 26.7103E

Elev: 2793

StdP: 13.27

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	11.8	-1.4	5.7	15.2	2.6	7.0	17.8	52.7	21.7	44.4	19.8	12.9	0	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	79.1	65.2	76.9	64.3	74.6	63.2	68.1	76.5	66.2	74.2	64.6	72.4	7.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.9	102.4	73.5	63.0	95.4	71.5	61.3	89.9	69.6	34.1	76.5	32.5	74.5	31.2	72.7	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
40.0	30.9	24.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	27.7	31.6	36.5	45.0	55.0	62.2	64.5	65.6	57.1	47.4	38.6	32.2
	DBStd	15.51	9.75	9.80	9.37	9.38	6.25	5.78	5.54	6.04	7.95	8.93	9.39	9.24
	HDD50	2968	690	518	428	203	26	1	1	0	27	159	361	554
	HDD65	6763	1155	936	884	600	311	120	77	70	255	545	793	1017
	CDD50	1885	0	2	9	54	181	366	450	483	240	80	18	2
	CDD65	206	0	0	0	1	2	35	62	88	18	0	0	0
	CDH74	610	0	0	0	1	12	79	197	275	46	0	0	0
	CDH80	43	0	0	0	0	0	1	20	20	2	0	0	0
Wind	WSAvg	9.7	10.7	11.3	11.1	10.9	8.5	9.3	8.3	8.3	9.2	9.8	8.8	10.6
Precipitation	PrecAvg	27.6	1.2	1.1	1.4	2.2	3.6	4.1	3.6	3.3	2.1	1.9	1.5	1.5
	PrecMax	36.9	2.8	3.1	2.6	4.3	7.3	7.1	7.9	6.6	4.1	6.6	3.0	3.5
	PrecMin	17.6	0.3	0.1	0.1	1.1	1.2	1.4	1.1	0.9	0.5	0.2	0.0	0.2
	PrecStd	5.5	0.7	0.7	0.7	1.0	1.7	1.5	1.8	1.5	1.1	1.4	0.8	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.8	58.8	63.2	70.1	74.7	79.1	82.3	82.4	79.4	68.8	62.8	57.4
		MCWB	43.3	46.0	47.8	54.1	57.7	67.0	67.8	68.0	63.6	54.9	51.1	43.3
	2%	DB	48.6	52.5	58.2	65.8	69.0	76.3	78.7	79.2	74.4	65.6	58.8	52.1
		MCWB	40.3	42.9	46.5	52.5	57.5	65.3	66.1	64.6	60.9	53.7	47.8	40.8
	5%	DB	45.9	48.2	53.5	62.6	66.5	73.8	75.8	77.1	70.9	63.0	54.9	47.7
		MCWB	39.2	40.6	43.4	51.0	56.5	64.0	64.7	63.8	58.3	52.3	47.0	40.1
	10%	DB	41.7	44.7	49.6	59.1	64.1	71.2	73.3	74.7	67.6	60.5	51.9	44.1
		MCWB	36.5	38.7	41.9	48.0	55.1	62.2	63.4	62.8	57.3	51.2	45.0	38.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.7	48.3	50.4	56.2	61.7	69.4	70.7	70.6	66.3	57.6	52.5	45.9
		MCDB	51.0	58.7	59.0	67.1	69.0	76.8	78.0	78.8	75.3	65.4	59.3	51.3
	2%	WB	41.7	43.9	47.4	53.9	59.2	66.8	67.9	68.0	62.0	55.4	50.6	43.1
		MCDB	47.7	51.1	56.2	63.7	66.2	74.3	76.2	76.6	71.2	62.8	56.7	50.5
	5%	WB	39.8	41.8	44.8	52.0	57.4	64.9	65.8	65.9	60.2	53.9	48.6	41.1
		MCDB	44.6	48.1	52.6	60.9	64.7	72.8	73.8	74.0	69.0	61.4	54.0	47.2
	10%	WB	37.0	39.3	42.6	50.0	55.8	63.2	64.2	64.2	58.4	52.1	46.6	39.0
		MCDB	41.5	43.9	48.9	58.4	62.7	70.9	71.3	72.3	66.8	58.7	51.2	44.3
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.0	11.6	12.1	11.8	12.0	11.9	12.3	10.9	10.2	10.1	9.4
		MCDBR	12.6	14.8	16.2	14.7	13.9	13.8	14.5	13.9	13.3	12.2	13.0	12.6
		MCWBR	8.1	8.7	8.7	9.0	8.5	9.3	8.7	8.0	8.2	8.0	7.4	7.3
	5% WB	MCDBR	11.4	13.1	15.3	13.3	13.0	13.3	12.8	13.8	12.4	11.3	10.8	11.7
		MCWBR	8.1	8.6	8.7	8.8	8.9	9.4	8.8	8.7	8.2	8.0	7.5	7.9
Clear-Sky Solar Irradiance	taub	0.291	0.316	0.349	0.401	0.381	0.382	0.396	0.407	0.371	0.340	0.316	0.292	
	taud	2.398	2.333	2.316	2.195	2.317	2.335	2.299	2.262	2.340	2.421	2.448	2.417	
	Ebn at Noon	260	272	278	272	281	281	275	267	267	261	248	246	
	Edh at Noon	23	30	35	43	39	39	40	40	34	27	22	21	
All-Sky Solar Radiation	RadAvg	410	670	1035	1413	1780	1886	1921	1737	1261	821	449	344	
	RadStd	49	78	108	145	146	144	144	121	113	93	42	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	-218	-292	+135	+49

Nomenclature: See separate page